

STUDY GUIDE – MIDTERM 1

Exam 1 will cover the following activities and learning goals:

ACTIVITY 1 - STATS, VISUALS, AND STORIES

- Define and identify observational units.
- Define and identify variables.
- Classify variables as categorical or quantitative.
- Extract basic information from simple data visualizations.
- Interact and manipulate variables within plots.

ACTIVITY 2 - BASICS OF DATA AND GRAPHS

- Interpret proportions as summary of categorical variables
- Identify and interpret visuals that are appropriate for categorical data (bar graphs).
- Describe the distribution of a categorical variable by referring to counts/proportions/percentages.
- Interpret measures of center (mean, median, mode).
- Calculate and interpret measures of variability (range, sd and IQR).
- Identify and interpret visuals that are appropriate for quantitative data (dotplots and histogram).
- Describe distribution of a quantitative variable by referring to shape, center, spread, and unusual observations.
- Compare distributions with respect to shape, center and variability between two or more distributions displayed in dotplots or histogram.
- Make predictions about centers and means, and about variability and standard deviations, based on the context of a variable.
- Develop critical statistical thinking

ACTIVITY 3 - SAMPLING METHODS

(SAMPLING METHODS: BIASED OR UNBIASED?)

- Describe and identify the population of interest.
- Collect and identify a sample.
- Identify the parameter of interest.
- Identify and calculate the statistic of interest.
- Identify biased and unbiased sampling methods based on the distribution of statistics.
- Identify tendency in bias (overestimate or underestimate the parameter of interest).
- Collect a simple random sample.
- Explain why increasing the sample size does not make a sample more representative of the population.

ACTIVITY 4 - SIMULATION AND PROBABILITY (GOATS & CARS)

- Conduct simulations to approximate probabilities.
- Identify the need to increase the number of repetitions in a simulation on the accuracy of an approximate probability.
- Identify and interpret probability as a long-run proportion.
- Select an appropriate decision involving a random process, based on probabilities and/or simulation results.
- Describe and identify the parameter of interest based on the description of a random process or population.

ACTIVITY 5 - ONE PROPORTION INTRODUCTION (CAN DOGS UNDERSTAND HUMAN CUES?)

- Implement the six steps of a statistical investigation given a particular statistical study
- Explain and apply the 3S Strategy (Statistic, Simulate, and Strength of evidence) when evaluating claims about a single population proportion using the count or sample proportion as the statistic
- Specify an appropriate chance model for a given set scenario" or "for a given set of assumptions
- Describe and conduct a simulation with many repetitions under the chance model by hand and using the One Proportion applet
- Interpret the results from a simulation
 - i. Identify what a dot on the plot is in the context of the study.
 - ii. Identify whether the observed statistic is unlikely under the assumption that the chance model is true.
 - iii. Identify the strength of the evidence against the chance model
 - iv. Identify whether the results of the study are unlikely to occur just by random chance.
- Use the results of the simulation to answer the research question in context.

ACTIVITY 6 - ONE PROPORTION SIMULATION (TASTING WATER)

- Conduct a simulation-based approach when evaluating claims about a single population proportion using the sample proportion as the statistic.
- Apply the correct symbols/words and distinguish between statistic and parameter when referring to a categorical variable
- State null and alternative hypothesis for a statistical test in words and symbols
- Calculate, interpret, and evaluate the p-value
- Use the p-value to answer the research question in context.

ACTIVITY 7 - ONE PROPORTION THEORY

(DO PEOPLE USE FACIAL PROTOTYPING?)

- Review important statistical ideas from previous class.
- Calculate and interpret the standardized statistic under a simulation-based approach
- Calculate and interpret the standardized statistic under a theory-based approach
- Conduct a theory-based approach when evaluating claims about a single population proportion.
- Identify whether the validity conditions for a one-sample z-test are met.
- Evaluate the strength of evidence against the null hypothesis based on the p-value and/or standardized statistic (z)
- State a conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value.
- Identify three factors and how they affect strength of evidence against the null hypothesis.

Below is the specific information that you need to know about each chapter of the textbook:

FROM THE PRELIMINARIES CHAPTER YOU SHOULD BE ABLE TO:

- Define probability in terms of long-run proportion
 - If asked to “define” probability don’t use the words probability, chance, likelihood, or odds or other synonyms
- Make sure you clearly indicate the random process being repeated and the outcome(s) of interest
- Identify the observational unit (the person or object for which we record information) for a study
 - To help identify the observational unit, think about the sample size/how many observations/pieces of data you have
- Define the variables (characteristic varying from observational unit to observational unit)
 - Make sure you can state these as variables for the specified observational unit (e.g., hair color or whether or not have red hair)
 - Don’t confuse with groups of people (red heads or those with black hair)
 - Make sure it’s clear there is more than one possible outcome (e.g., not just “red hair”)
 - Don’t confuse with “results” (the proportion of students with red hair)
 - Don’t confuse with research question (are there fewer red heads among males?)
 - Try to specify as a question we can “ask” **each** observational unit (what is your hair color?)
- Classify the variable type as quantitative or categorical
- Describe the behavior of a distribution of a quantitative variable (shape, center, variability, unusual observations) in context and suggest explanations for unusual behavior

FROM CHAPTER 1 YOU SHOULD BE ABLE TO

- Interpret a bar graph
- Identify the sample size (n)
- Distinguish between parameter (numerical summary of population or process) and statistic (numerical summary of sample)
 - Be able to use appropriate symbols to refer to these
 - Differentiate between proportion and percentage
 - Define a parameter or a statistic (in words and/or recognize the value)
- Understand the concept of simulation (replicating a hypothetical study under specific conditions)
- Describe how to use the One Proportion applet to carry out a simulation of n trials with probability of success (π)
 - Be able to indicate the input values
 - Map out the simulation parallels to the original study
 - Identify and interpret what each dot in the null distribution represents. Make sure you include the following details
 - Context of the study
 - The statistic that is being plotted
 - The sample size
 - Be able to anticipate some aspects of the behavior of the null distribution (e.g., where it will be centered)
 - Use the output to describe typical and atypical values for the statistic when the null hypothesis is true
 - Be able to produce an estimated p-value
- Also be able to roughly approximate the p-value yourself based on the graph results (is the statistic in the tail of the distribution so small p-value)
- Be able to evaluate the strength of evidence against the null hypothesis based on the value of the p-value.
- Differentiate evaluation based on the p-value from interpretation of the p-value.
- When interpreting the p-value, make sure you refer to the following
 - Context
 - Statistic
 - Direction
 - Assumption
- Use the estimated p-value to make a conclusion about the research conjecture (vs. the by chance alone hypothesis)
 - Be able to explain the logic/reasoning behind the decision (e.g., is it likely to have obtained our statistic by chance alone)
- State null and alternative hypotheses about a the population parameter
 - Using symbols and/or words
 - Including a “less than” alternative, a “greater then” alternative, and a “not equal to” alternative, based on the research question
 - Understand idea of two-sided p-value considering evidence in either direction
- Decide whether results are unlikely to have happened by chance.
 - Decide whether to reject or fail to reject the null hypothesis

- Standardizing an observation (observation-mean)/SD_{NULL} as a way to measure distance and how far in the tail of the distribution the observation is
 - Interpreting as number of standard deviations above or below the mean (positive or negative values)
 - Considering values more than 2 or 3 standard deviations from the mean as “in the tail” of the distribution. For instance, a z value of 2 typically corresponds to a two-sided p-value below .05
- Explain factors that impact strength of evidence (sample size, distance between observed and hypothesized, one-sided vs. two-sided)
- Apply the theory-based approach for **one proportion** (one-sample z-test)
 - Calculate and interpret the theoretical standard deviation of sample proportions:

$$\sqrt{\frac{\pi(1-\pi)}{n}}$$
 - State and check the validity conditions for the normal approximation to the null distribution
 - Use technology (**Theory Based Inference** applet) to determine the theory-based p-value

FROM CHAPTER 2 YOU SHOULD BE ABLE TO

- Identify the population of interest in a research study
- Identify and distinguish between the population (entire group of observational units of interest) vs. sample (those observational units from the population we record information for)
 - Parameter as a numerical summary of a population
 - Symbols for statistics vs. parameters
- Identify the cause and direction of bias in a biased sampling method
 - How can you tell whether a sampling method is biased from a distribution of sample statistics?
- Be able to identify/describe a sampling frame
- Discuss how to use a sampling frame to select a simple random sample (SRS)
 - Specify a sampling frame in a particular context (i.e., the physical list of members of population like the phone book)
 - Number the units in the frame, use a random number mechanism to generate ID numbers, report the members of the sample
- Explain the purpose of random sampling (produce a sample representative of population)
 - Long-run mean of sample statistics will be equal to the population parameter
 - Allows you to generalize conclusions (significant or not!) from the sample to the population.
- Discuss how the amount of “random sampling error” depends on the sample size
 - Larger samples are more precise = statistics cluster more closely around the parameter
*Note that this does **not** depend on the population size as long as the population size is much, much larger than the sample size (10x)*
- Interpret a histogram or dotplot/ compare distributions of quantitative data
 - Eyeball the center and spread of the distribution, identify shape as roughly symmetric or not, skewed left vs. skewed right
 - Interpret mean as a measure of center (include measurement units).

- Use the word “typical” when interpreting the mean or median in context
 - Interpret standard deviation as a measure of spread (include measurement units)
 - Be able to compare (or conjecture) how the spread of two distributions compares from the graphs/nature of variables involved
 - Always put your comments into the context of the study
- Decide which graphs (bar graph vs. dotplot/histogram) to use with which type of variable (categorical vs. quantitative)
- **NOT NEEDED FOR THIS MIDTERM** Assess the statistical significance of an observed sample mean based on the null distribution
 - ~~State null and alternative hypotheses about μ~~
 - ~~Explain what determines the shape, center, and variability of the null distribution of sample means~~
 - ~~Calculate the standard error of the sample mean $SE(\bar{x}) = \frac{s}{\sqrt{n}}$~~
 - ~~“Standard error” just means it is an estimate of a theoretical standard deviation. You can pretty much ignore this distinction and just think in terms of standard deviation.~~
 - ~~Calculate and interpret the standardized statistic (t-statistic)~~
 - ~~Determine whether a distribution of sample means is expected to be normal or approximately normally distributed~~
 - ~~If validity conditions are met, can use **Theory-Based Inference** applet to obtain p-value~~
- **NOT NEEDED FOR THIS MIDTERM** Define Type I and Type II errors in context
- **NOT NEEDED FOR THIS MIDTERM** Identify when you could commit a Type I error.
- **NOT NEEDED FOR THIS MIDTERM** Identify when you could commit a Type II error.